Rahul **Satish**

Ph.D. Researcher, ITU Copenhagen

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Education

Present | **Department of Computer Science, IT University of Copenhagen** | **Copenhagen, Denmark**
Jan 2024 | Ph.D. in Computer Science (Expected to submit in March 2027) | Advisor: Prof. Bernardo David
Thesis: Efficient Dynamic MPC Protocols

Nov 2023 | **Alexander Kofkin Faculty of Engineering, Bar Ilan University** | **Ramat Gan, Israel**
Sep 2021 | M.Sc. in Electrical Engineering (Computer Engineering Track) | Advisor: Prof. Carmit Hazay
Thesis: Circumventing Lower Bounds in Garbled Circuits

Aug 2021 | **Birla Institute of Technology and Science (BITS) Pilani** | **Goa, India**
Aug 2016 | Dual Degree Programme, M.Sc.(Mathematics) & B.E.(Electronics & Instrumentation)
Project: Exploring Round Optimal Solutions for the PSI-Sum with Cardinality Problem

Experience

Dec 2026 | **Information systems Lab, Lucerne University of Applied Sciences and Arts** [🌐] **Lucern, Switzerland**
Sept 2026 | *Ph.D. Visiting Researcher | Advisor: Prof. Chen-Da Liu-Zhang*
Visiting the Information Systems Lab during my Ph.D.

Feb 2026 | **Social Informatics Laboratories, NTT Musashino R&D Center** [🌐] **Tokyo, Japan**
Nov 2025 | *Ph.D. Intern | Advisor: Prof. Masayuki Abe*
Interning at NTT R&D, Social Informatics Laboratories

Oct 2021 | **Ethereum Foundation | Privacy and Scaling Explorations Group** [🌐] **Remote**
Jul 2021 | *Consultant Researcher*
Worked with a team developing a zk-rollup that directly supports the Ethereum Virtual Machine(EVM)

Dec 2021 | **Chain Reaction | Cryptography Research Group** [🌐] **Remote**
Jul 2021 | *Consultant Researcher*
Worked on modelling and understanding existing levelled and fully Homomorphic Encryption Schemes

Jun 2021 | **QED-it Systems | Cryptography Group** [🌐] **Remote**
Jan 2021 | *Research Intern* | Advisors: Daniel Benarroch, Michael Adjedj
Worked on constructing efficient circuits for the BFV scheme, for comparing & sorting encrypted integers.

Aug 2020 | **IBM Research | Blockchain and Smart Contracts Group** [🌐] **Bangalore, India**
May 2020 | *Research Intern* | Advisors: Dr. Dhinakaran Vinayagamurthy, Nitin Singh
Designed a modular framework to write interactive ZKP systems on arithmetic circuits

Publications*

S=In Submission, C=Conference, W=Workshop, P=Poster/Demo, J=Journal | [DBLP], [Scholar]

[S.4] **(A)synchronous Publicly Verifiable Broadcast Optimal DKG with Cheater Identification**
Bernardo David, Rahul Satish
[**In Submission**]

[S.3] **Dual Adaptive Security for Garbling Schemes**
Anasuya Acharya, Carmit Hazay, Rahul Satish
[**In Submission**]

[S.2] **Oblivious Transfer Extension for Dynamically-Scheduled Secure Computation**
Masayuki Abe, Peihao Li, Miyako Ohkubo, Octavio Perez-Kempner, Rahul Satish, Mehdi Tibouchi
[**In Submission**]

[S.1] **Eras MPC: Granular Preprocessing and Abort Recovery with Dynamic Committees** [📥]
Bernardo David, Ivan Damgard, Rahul Satish
[**In Submission**]

*indicates authors listed in alphabetical order following the Hardy-Littlewood rule.

[C.4] **Sublinear-Communication Layered MPC from HSS** [📦]
Anasuya Acharya, Pierre Meyer, Divya Ravi, Rahul Satish
International Conference on the Theory and Application of Cryptology and Information Security **[ASIACRYPT'26]**

[C.3] **Oblivious Garbling and its Applications** [📦], [🔗]
Tomer Ashur, Carmit Hazay, Rahul Satish
International Conference on Security and Cryptography for Networks **[SCN'26]**

[C.2] **The Landscape of Reusable Garbling** [📦], [🔗]
Anasuya Acharya, Carmit Hazay, Rahul Satish
International Conference on Security and Cryptography for Networks **[SCN'26]**

[C.1] **Rumors MPC: GOD for Dynamic Committees, Low Communication via Constant-Round Chat** [📹], [📦], [🔗]
Bernardo David, Arup Mondal, Rahul Satish
International Conference on the Theory and Application of Cryptology and Information Security **[ASIACRYPT'25]**

[W.1] **Garbling 3-input AND gates** [📹], [🔗]
Tomer Ashur, Carmit Hazay, Rahul Satish
Conference for Failed Approaches and Insightful Losses in Cryptology **[CFAIL'23]**

Talks

"Sublinear-Communication from Layered MPC from HSS"

› TPMPC 2026 [🌐] [📹] May 2026 (Aarhus, Denmark)

" Rumors MPC: GOD for Dynamic Committees, Low Communication via Constant-Round Chat"

› Asiacrypt 2025 [🌐] [📹] December 2025 (Melbourne, Australia)

"Garbling gadgets and its applications to oblivious garbling"

› TPMPC 2024 [🌐] [📹] June 2024 (Darmstadt, Germany)

"Garbling 3-input AND gates"

› CFAIL 2023 [🌐] [📹] August 2023 (Remote)

Select Technical Projects

Generating custom polynomial constraints for efficient PLONK circuits Jun'21 - Oct'21

› Working on a zk-rollup that directly supports the Ethereum Virtual Machine(EVM). Prior to that, we were developing a ZKP layer over the EVM to be able to validate blocks through the PLONK proving system.

Efficient sorting and comparison in the homomorphic setting Jan'21 - Jun'21
Advisors: Daniel Benarroch, Michael Adjedj [🌐 Blog]

› Identifying and implementing the best possible ways to construct HE circuits for comparison and sorting integer data.

Non-Interactive Proof Generation from Interactive Zero Knowledge Protocols May'20 - Aug'20
Advisors: Dr. Dhinakaran Vinayagamurthy, Nitin Singh [🌐 Repository]

› Designed a modular framework for Interactive Zero Knowledge Protocols which was used to convert it to a non-interactive protocol. Implemented additional features for the design to support oracles, protocol composition, etc.. and tested existing protocols like Ligero on it.

Academic Service

External Reviewer AC'26, Crypto'26, PKC'26, EC'26, PKC'25, AC'25, Crypto'25, PKC'25, ICDCS'25, TCC'24, EC'24, DCC'24, JoC'23, SAC'22 , EC'22, CCS'21, TCC'21

Teaching and Leadership Roles

Introduction to Cryptography, Cryptographic Computation and Blockchains *Courses Instructor* at ITU

› Responsibilities included evaluating tutorials, and helping students with the coursework and home assignments.

IEEE ANTS, 2019, BITS Goa *Web Development Lead* [Website] May'19 - Dec'19

› Responsible for building and maintaining website of the IEEE International Conference on ANTS 2019.

Introduction to Applied Cryptography, Quark Summer Technical Project *Instructor* May'19 - Aug'19

- Voluntarily mentoring students from across the nation to get used to cryptographic objects and the math behind it.

BITSkrieg, Ethical Hacking and PenTesting Club, BITS Goa *Co-ordinator* Aug'18 - Jun'19

- Responsibilities include ensuring that the club works towards achieving its annually laid out goals.
- Handled a lecture series on Applied Cryptography for students of the college, being a member of the club.
- Involved in solving Capture the Flag(CTF) events such as Google CTF, PICO CTF, DEFCON regularly.

Quark Controls, BITS Goa *Core Member* Mar'17 - Feb'18

- Successfully established contacts with companies like Mozilla, JP Morgan, LIGO for hosting workshops on campus.
- Responsible in helping the team arrange seminars and talks for the students on technical issues.

References

- Prof. Bernardo David ..... *Associate Professor, IT University of Copenhagen, Denmark* [🌐][✉]
- Prof. Carmit Hazay ..... *Professor, Bar Ilan Univeristy, Israel* [🌐][✉]
- Prof. Masayuki Abe ..... *Senior Distinguished Researcher, NTT Social Informatics Laboratories, Japan* [🌐][✉]
- Dr. Tomer Ashur ..... *Co-Founder, 3MI Labs, Belgium* [🌐][✉]
- Daniel Benarroch ..... *CEO, Inversed Tech, Spain* [🌐][✉]